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Grounding Fine-Grained Multimodal Captions: A Pipeline for Integrating PixMo-Cap into HyCoCLIP

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Abstract

Capturing the complex semantics of a visual scene through natural language requires recognizing the hierarchical relationships between spatial regions within an image. While modern contrastive models excel at global image-text retrieval, fine-grained spatial details are frequently lost when mapping detailed descriptions into standard Euclidean spaces. Recent work has highlighted the promise of hyperbolic representations for vision-language understanding, as hyperbolic space naturally models hierarchical structures. To bridge the gap from Euclidean space to hyperbolic space, in our work, we present a grounding pipeline that takes advantage of PixMo-Cap's dense captioning potential within the hyperbolic architecture of HyCoCLIP [10]. Building on the grounding approach originally used to prepare the GRIT dataset [11] for HyCoCLIP's training, our pipeline expands base noun phrases into rich referring expressions via dependency tree traversal. However, to handle the structural complexity of dense narrative text, which in some captions contain meta descriptors of the scene and add commentary, we enhance this process with single-sentence context isolation, multi-layer filtering to strip abstract framing and subjective modifiers, and strict clause-boundary truncation. We assess the resulting dataset through statistical characterization of the extracted regions and phrases, together with a manual, sample-level quality review, indicating that this clean entity extraction meaningfully reduces text-side noise and yields more consistent, well-localized region-phrase pairs suitable for training HyCoCLIP.

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Chapter 1

Introduction

1.1 The Grounding Problem Behind Hyperbolic Vision-Language Models

Vision-language models that are trained with contrastive objectives, such as CLIP [1], have become the main underlying principle behind a large fraction of modern multi-modal AI systems for zero-shot classification, and the visual encoders behind many text-to-image and captioning pipelines. Their success rests on aligning an image and its caption in a shared embedding space by pulling matching pairs together and pushing mismatched ones apart, learned from hundreds of millions of noisy image–text pairs. This concept is remarkably efficient in capturing what is depicted in an image as a whole, while it is not sufficient in capturing how the components of a scene relate in the hierarchical and compositional sense. The contrastive objective is limited to comparing an image itself to a whole caption only, and it is well documented that CLIP-style embeddings struggle with tasks that require binding attributes and relations to specific entities rather than to the image as a whole. A more recent line of work has challenged this limitation by changing the geometry of the embedding space of traditional vision-language models. To exemplify, HyCoCLIP [10] replaces the Euclidean embedding space used by CLIP with hyperbolic space. The exponential growth in volume as a function of radius in a hyperbolic space makes it naturally suited to represent hierarchical, part vs. whole relationships. As illustrated in Figure 1.1, an image, its contained objects, and the descriptors for each can be placed at different depths in the space, increasing the specificity of concepts further outward. To learn this hierarchy, however, the model needs more than the image itself and a single global caption. A nuance appears where supervision is required for the process of linking regions of an image to the phrases that describe them, which is a property provided by datasets such as GRIT [11].

This is where the motivation for the presented work begins. GRIT-style grounding data, despite being well suited to HyCoCLIP's training scheme, is limited in scale and in the richness of the descriptions. The captions are typically short, noun phrases rather than full, descriptive sentences, restricting the information the model can ultimately learn. PixMo-Cap [22], on the other hand, offers image captions enriched via human annotation. Because these captions are transcribed from detailed spoken descriptions provided by human annotators, the resulting caption is longer, densely descriptive, while also sometimes narrative. These narrations mention many meta-descriptors, such as entities, attributes, and spatial relations per image, as shown in Figure 1.2. In theory, this makes PixMo-Cap a much more informative

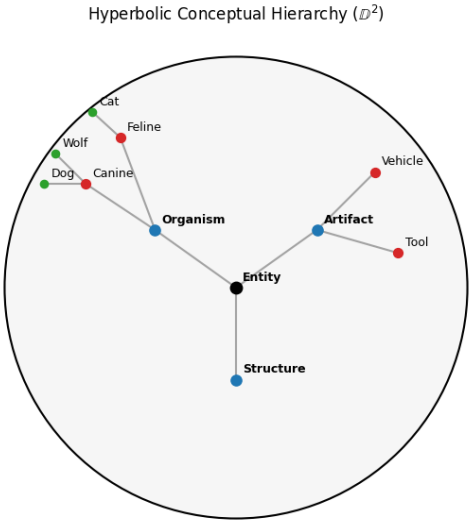


Figure 1.1. Example of hyperbolic hierarchy.

source of supervision than GRIT. However, in practice, this richness is exactly what makes PixMo-Cap complicated to use for the kind of grounding HyCoCLIP requires. Unlike GRIT's short, isolated noun phrases, PixMo-Cap's captions are long, free-form paragraphs, mixing concrete visual descriptions with narrative asides, references across sentences, and meta-descriptor phrases with no localizable correspondence in the image. Turning this kind of text into the image-text pairs that a grounding-based training scheme requires therefore creates a non-trivial extraction and grounding problem.

The goal of this thesis is to address this problem by implementing a pipeline that converts PixMo-Cap's dense, narration-style captions into fine-grained, region-grounded image–text supervision suitable for training HyCoCLIP, and finally, to assess the quality of the resulting data over certain metrics. This involves extracting meaningful, visually groundable phrases, localizing those phrases in the corresponding images, and filtering the result into a clean training-ready dataset. This ensures that the abstract phrases that cannot be grounded and noisy detections are filtered out, before converting captions whose length and narrative structure differ into the short-caption data that can be utilized in the hyperbolic space representation. Thus, our primary question can be summarized as follows: "Can a dense, narration-derived caption dataset such as PixMo-Cap be systematically transformed into supervision for HyCoCLIP, a hyperbolic contrastive model?"

The remainder of this thesis is organized as follows. Chapter 2 introduces the background needed to follow the rest of the work, including contrastive vision-language pretraining and CLIP, the hyperbolic embedding space used by HyCoCLIP, the PixMo-Cap dataset and its role in grounded contrastive training, and the caption component of PixMo-Cap. Chapter 3 describes the methodology developed to bridge PixMo-Cap and HyCoCLIP, including the preprocessing pipeline, the difficulties posed by PixMo-Cap's caption format relative to GRIT, the discussion of alternative approaches, and the final design that was adopted. Chapter 4 presents the metrics used to evaluate the pipeline and discusses what they signify. Finally, Chapter 5 summarizes the main findings of the thesis and outlines directions for future work, emphasizing how this data pipeline serves as a stepping stone toward broader



"This nighttime image captures a dynamic point of view from a motorcycle rider on a dimly lit street, giving a sense of motion and speed. The viewer sees the motorcycle's handlebars and front dash, including a prominent circular speedometer and a smaller gas gauge, with a key inserted in the ignition, attached to a brown leather key holder. The handlebar details and top of the seat are slightly cut off due to the narrow framing. The black body of the vintage motorcycle is evident. The road is gray asphalt, bordered by white lines, with several streetlights illuminating the right side. Ahead, the streetlights cast a soft glow on the path forward, with a blurry outline of another motorcycle visible further down the road. To the left, partially fuzzy buildings with a white and blue sign are visible, while the right side of the road features green grass and a touch of green foliage on the left."

Figure 1.2. Example of an image and its dense caption from PixMo-Cap.

extensions of grounded hyperbolic models.

Chapter 2

Background

2.1 Contrastive Multi-modal Learning

Learning joint representations of images and text has become a foundational architecture in modern computer vision, replacing task-specific supervised pipelines with a single general-purpose training objective. Visual encoders behind modern text-to-image and captioning pipelines [1, 3] project modalities into a shared space. In this architecture, an image encoder and a text encoder are trained together so that the representations of a matching image-text pair are pulled close together in a shared embedding space, while representations of mismatched pairs are pushed apart. Once trained, the resulting embeddings support zero-shot classification, cross-modal retrieval, and transfer to a wide range of downstream tasks without task-specific fine-tuning. Evaluation in the model generally mirrors the training objective itself, as models are scored on how well they retrieve the correct caption for a given image (or vice versa) and on zero-shot classification accuracy across benchmark datasets, rather than through supervised metrics tied to a fixed label set.

The foundational advancements in multi-modal architectures, especially in the language-image category, have been brought by dual-encoder architectures, such as CLIP (Contrastive Language-Image Pre-training) [1] and ALIGN [3]. In principle, these models process images I and full caption texts T via independent visual and text encoders. Then, these encodings are mapped into a shared d-dimensional Euclidean space $\mathbb{R}^d$ and normalized onto a hyper-sphere $\mathbb{S}^{d-1}$, where semantic similarity is captured through cosine distance. Formally, given an image I and caption T, the visual and text encoders produce feature vectors

$$v = f_I(I) \in \mathbb{R}^{d_v}, \qquad u = f_T(T) \in \mathbb{R}^{d_t}, \tag{2.1}$$

which are projected into the shared d-dimensional space via learned linear maps $W_I \in \mathbb{R}^{d \times d_v}$ and $W_T \in \mathbb{R}^{d \times d_t}$:

$$x = W_I v, \qquad y = W_T u. \tag{2.2}$$

Both embeddings are then ℓ_2-normalized onto the unit hypersphere $\mathbb{S}^{d-1}$:

$$\hat{x} = \frac{x}{\|x\|_2}, \qquad \hat{y} = \frac{y}{\|y\|_2}. \tag{2.3}$$

On the hypersphere, cosine similarity reduces to a simple dot product:

$$\text{sim}(\hat{x}, \hat{y}) = \hat{x}^\top \hat{y} = \cos(\theta). \tag{2.4}$$

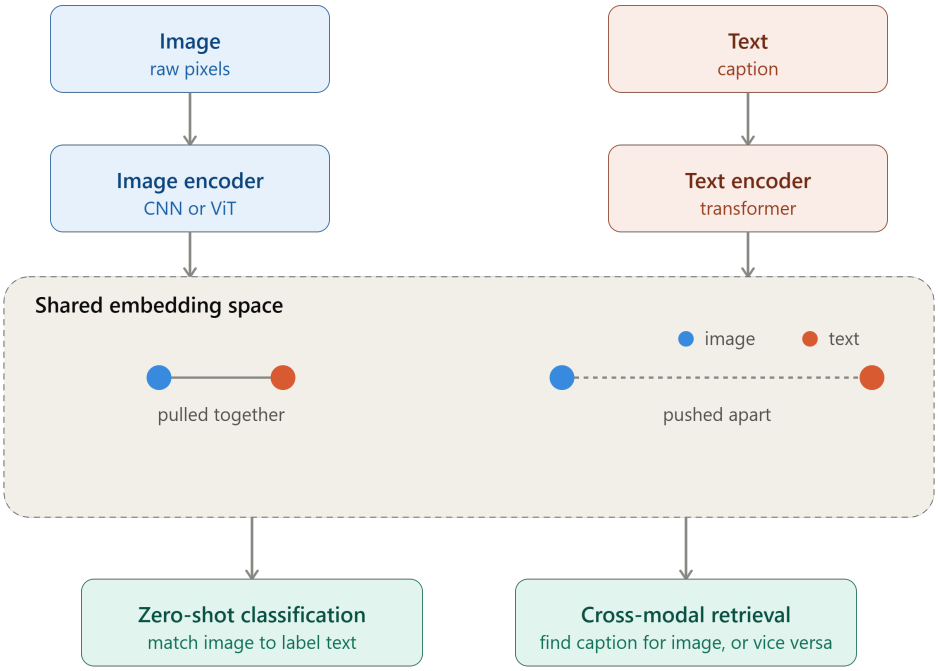


Figure 2.1. Contrastive image-text embedding architecture: An image encoder and text encoder map their inputs into a shared embedding space, where matching image-caption pairs are pulled together and mismatched pairs are pushed apart. The resulting embeddings support zero-shot classification and cross-modal retrieval.

Given a batch of N matched image-text pairs $\{(\hat{x}_i, \hat{y}_i)\}_{i=1}^{N}$ and a learnable temperature τ, the model is trained with a symmetric InfoNCE objective [4, 1]:

$$\mathcal{L}_{I\to T} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\hat{x}_i^\top \hat{y}_i/\tau)}{\sum_{j=1}^{N}\exp(\hat{x}_i^\top \hat{y}_j/\tau)}, \tag{2.5}$$

$$\mathcal{L}_{T\to I} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\hat{y}_i^\top \hat{x}_i/\tau)}{\sum_{j=1}^{N}\exp(\hat{y}_i^\top \hat{x}_j/\tau)}, \tag{2.6}$$

$$\mathcal{L} = \frac{1}{2}\left(\mathcal{L}_{I\to T} + \mathcal{L}_{T\to I}\right). \tag{2.7}$$

CLIP is an instance of this architecture that demonstrates that a contrastive objective, which is trained on hundreds of millions of noisy web-scraped image-text pairs, could rival supervised models on a wide range of visual recognition tasks without being trained on those tasks directly. Despite its success, this design has a structural limitation. It is well documented that CLIP-style embeddings struggle with binding attributes and relations to the correct entities [5, 6]. Because CLIP treats an entire image and an entire caption as a single, holistic pair, it has no explicit mechanism for representing the internal structure of a scene, namely the contained individual objects, their spatial relationships, or the part-whole relationships between an object and the larger image it belongs to. This manifests itself as weak compositional and

relational reasoning in CLIP-style models, where it is apparent in specific tasks, such as distinguishing "a dog chasing a cat" from "a cat chasing a dog" [5, 6], since both captions activate the same bag of visual concepts. This limitation motivates a reconsideration not of the contrastive objective itself, but of the geometric space in which these embeddings live.

2.2 Hyperbolic Representation Learning and HyCoCLIP

Standard vision-language embeddings live in Euclidean space, where all directions are geometrically equivalent, and the volume is polynomial with respect to radius. This geometry is argued to be a poor fit for hierarchical data [7, 9], where a small number of general, high-level concepts (e.g., "animal") entail a much larger number of specific instances (e.g., "labrador", "tabby cat", "sparrow"). A hyperbolic space, most commonly modeled via the Poincaré ball or Lorentz model, has constant negative curvature, causing volume to grow exponentially with radius [7]. This property makes it naturally suited to embedding tree-like or hierarchical structures with low distortion where general concepts can be placed near the origin, while increasingly specific concepts are pushed toward the boundary of the ball [7, 8], with the property that a more general concept's embedding lies on the geodesic path toward all of its more specific descendants [8]. As a result, this kind of hierarchy diffuses throughout vision-language data: "dog" entails "animal", a cropped region of an image entails the full image it was cropped from, and a short phrase entails the fuller sentence describing the same scene.

MERU [9] was the first to bring this geometry to contrastive vision-language pretraining, embedding images and text on a hyperboloid rather than a hypersphere and showing that the resulting space is more interpretable in terms of successive crops of the same image tracing a path toward the origin, as the concepts become more general. However, MERU still treats each image and caption as an undifferentiated whole, capturing hierarchy only *across* image-text pairs, not *within* a single image or caption in a compositional sense.

HyCoCLIP (Hyperbolic Compositional CLIP) [10] extends this line of work by explicitly modeling hierarchy at the sub-image and sub-caption level. In addition to the whole image and its full caption, HyCoCLIP incorporates image "boxes" which correspond to cropped regions consisting of individual objects and text "boxes" that are basically noun phrases or spans describing those objects. HyCoCLIP organizes these four elements (image, image box, text, text box) into a single hierarchical structure. This structure captures both an *inter-modal* hierarchy, in which text is treated as more general than the image it describes, and two *intra-modal* hierarchies, in which a cropped box is treated as more specific than the full image (and a phrase as more specific than the full caption) it was drawn from. HyCoCLIP optimizes this structure jointly through a hyperbolic contrastive loss and an entailment-cone loss [8, 10], the latter explicitly encouraging general concepts to be embedded within the "cone" of their more specific descendants. HyCoCLIP was originally trained and evaluated on the GRIT dataset [11], whose region-grounded structure supplies exactly the image-box/text-box hierarchy the model requires, which is also a dependency that motivates the next section. Formally, following [9, 10], HyCoCLIP embeds all representations in the Lorentz model of hyperbolic space

$$\mathbb{L}^n = \left\{ \mathbf{p} \in \mathbb{R}^{n+1} : \langle \mathbf{p}, \mathbf{p} \rangle_{\mathbb{L}} = -\frac{1}{\kappa},\ p_0 = \sqrt{1/\kappa + \|\tilde{\mathbf{p}}\|^2},\ \kappa > 0 \right\}, \tag{2.8}$$

where $-\kappa$ is the curvature, p_0 is the time-axis coordinate, and $\tilde{\mathbf{p}} \in \mathbb{R}^n$ the spatial

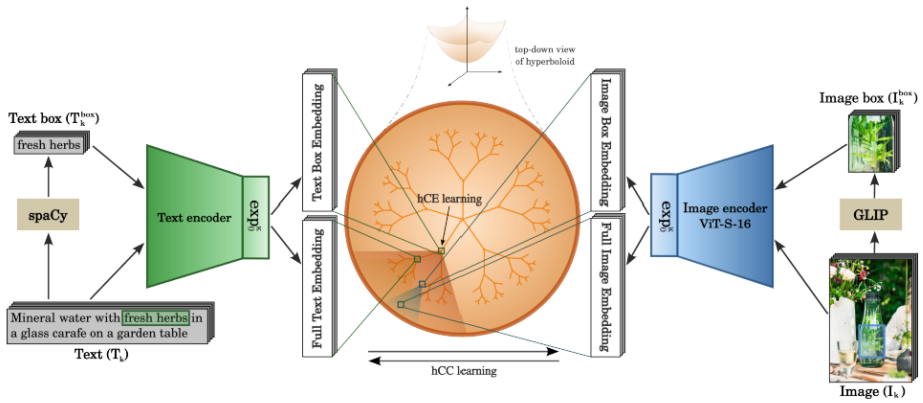


Figure 2.2. An overview of HyCoCLIP. Extracted Text and image boxes are embedded by HyCoCLIP's encoder modules, projecting the representations in the hyperbolic latent space, while also preserving the inter-modal and intra-modal relationships by locating them according to their broadness and by using entailment cones to give an interpretable structure to the learned latent space.

coordinates, with Lorentzian inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle_{\mathbb{L}} = -p_0 q_0 + \langle \tilde{\mathbf{p}}, \tilde{\mathbf{q}} \rangle_{\mathbb{E}}. \tag{2.9}$$

The geodesic distance between two points on $\mathbb{L}^n$ is

$$d_{\mathbb{L}}(\mathbf{p}, \mathbf{q}) = \sqrt{1/\kappa} \cdot \cosh^{-1}\big(-\kappa \langle \mathbf{p}, \mathbf{q} \rangle_{\mathbb{L}} \big). \tag{2.10}$$

Euclidean encoder outputs are mapped onto the hyperboloid via the exponential map at a point $\mathbf{p}$, for a tangent vector $\mathbf{v} \in T_{\mathbf{p}}\mathbb{L}^n$:

$$\exp_{\mathbf{p}}^{\kappa}(\mathbf{v}) = \cosh(\sqrt{\kappa}\|\mathbf{v}\|_{\mathbb{L}})\, \mathbf{p} + \frac{\sinh(\sqrt{\kappa}\|\mathbf{v}\|_{\mathbb{L}})}{\sqrt{\kappa}\|\mathbf{v}\|_{\mathbb{L}}}\, \mathbf{v}, \tag{2.11}$$

with encoder features treated as tangent vectors at the origin $\mathbf{0} = (\sqrt{1/\kappa}, 0, \ldots, 0)^\top$ and projected via $\exp_{\mathbf{0}}^{\kappa}$.

Given a dataset $D = \{(I_k, T_k)\}_{k=1}^{K}$ of image-caption pairs, each augmented with an image box I_k^{box} and text box T_k^{box}, HyCoCLIP jointly optimizes a hierarchical Compositional Contrastive (hCC) loss over all four elements using Lorentzian-distance-based similarity, and a hierarchical Compositional Entailment (hCE) loss that constrains the exterior angle $\phi(\mathbf{p}, \mathbf{q})$ between a specific concept $\mathbf{q}$ and its general parent $\mathbf{p}$ to lie within an aperture threshold $\eta\, \omega(\mathbf{p})$, enforced over the pairs (I, T), $(I^{\text{box}}, T^{\text{box}})$, (I, I^{box}), and (T, T^{box}). The overall hierarchical Compositional (hC) objective is a weighted sum of the two:

$$\mathcal{L}_{\text{hC}} = \mathcal{L}_{\text{hCC}} + \lambda\, \mathcal{L}_{\text{hCE}}. \tag{2.12}$$

2.3 Grounded and Region-Level Image-Text Data

The hierarchical training signal HyCoCLIP relies on does not occur naturally in ordinary image-caption datasets, which associate a singular caption with an entire

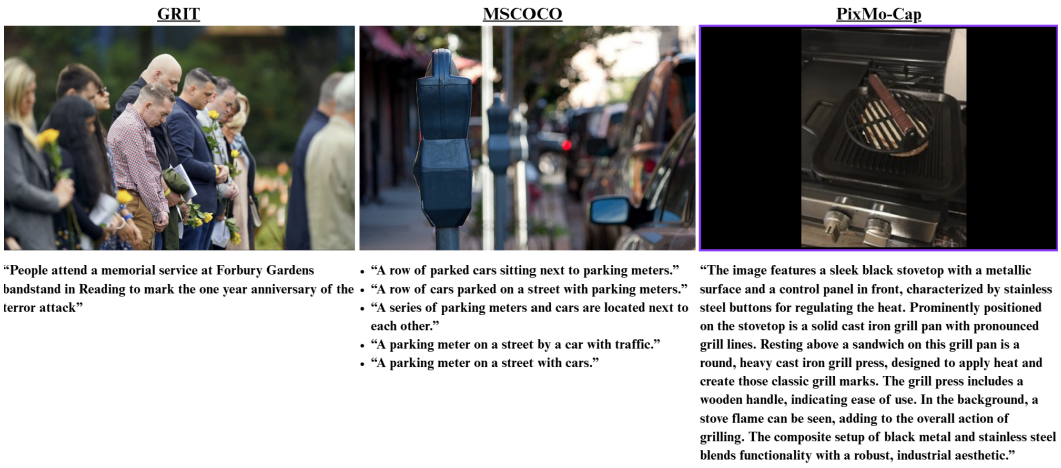


Figure 2.3. Sampled image captions from GRIT, MS-COCO, and PixMo-Cap datasets.

image and do not give information on which words correspond to which regions. GRIT (Grounded Image-Text pairs) [11], introduced alongside Kosmos-2, was constructed specifically to supply this missing structure. Starting from image-caption pairs drawn from existing web-scraped data, GRIT's construction pipeline first extracts noun phrases from each caption using a dependency parser, associates each noun phrase with a corresponding image region via a pretrained detector, and then expands these noun-chunk-region pairs into fuller referring expressions by recursively traversing the caption's dependency tree. The resulting dataset contains roughly 91 million images linked to more than 137 million bounding boxes and 115 million associated text segments [11], which is substantially larger than earlier grounding datasets, such as RefCOCO [12], RefCOCO+ [13], RefCOCOg [14], and Visual Genome [15]. Because each image arrives already decomposed into region-phrase pairs, GRIT provides HyCoCLIP conveniently with the image-box/text-box hierarchy its training objective is built around, without requiring any other additional annotation step.

Producing this kind of grounded data at scale depends on open-vocabulary object detection, and grounding models capable of localizing any arbitrary free-form textual descriptions rather than a small fixed set of trained categories. This represents a departure from earlier closed-set detectors such as Faster R-CNN [16] and YOLO [17], which can only localize object classes seen during training, and therefore cannot support the unrestricted vocabulary of naturally occurring captions. Transformer-based open-set detectors such as GLIP [18] and Grounding DINO [19], instead, reformulate detection as a vision-language matching problem: an image encoder and a text encoder produce joint representations, and the model is trained to align image regions with the specific words or phrases that describe them, enabling it to localize objects described by any natural-language prompt at inference time. It is precisely this capability to localize image boxes based on text that makes constructing datasets like GRIT feasible in the first place, and the same kind of tooling is directly relevant to any other pipeline that must derive region-level segmentation from densely captioned but otherwise "raw" data, as GRIT's own dependency-parsing and phrase-expansion approach [11] forms the starting point for the pipeline developed in Chapter 3.

2.4 Dense Captioning Datasets and PixMo-Cap

Datasets that avail image captioning have grown steadily richer over time, moving from providing short, templated descriptions toward longer, more detailed human annotations. MS-COCO [20], one of the most widely used captioning benchmarks [21], pairs each image with five short, independently written captions, typically about ten words long and focused on the most salient aspect of the scene. While sufficient for basic scene recognition, such captions' detail-capturing capacity is little, its background elements, or the relationships between multiple objects within it gets lost, which is precisely the kind of information that hierarchical, compositional training objectives depend on.

PixMo-Cap [22], introduced as part of the PixMo data suite for training the Molmo family of vision-language models, was designed explicitly to close this density gap. Rather than asking annotators to type texts that are specifically constrained as captions, annotators were instead asked to *describe* each image aloud for 60 to 90 seconds [22]. These spoken descriptions were transcribed via automatic speech-to-text and subsequently summarized or cleaned by a language-only large language model into a single polished caption for each image, yielding 712,000 images paired with 1.3 million transcripts and captions [22]. The resulting captions averaged around 196 words [22], which is more than one order of magnitude longer than MS-COCO's roughly ten-word captions [21].

This density of information makes PixMo-Cap an attractive source of rich, non-distilled textual supervision. However, unlike GRIT [11], PixMo-Cap's captions are not linked to specific image regions by default: each caption describes the image as a continuous, free-form narration rather than as a structured set of region-phrase pairs. This creates a need for a bridging tool with respect to HyCoCLIP's data requirements, which is satisfying the model's compositional training objective depending on exactly the kind of image-box/text-box hierarchy that GRIT provides by construction. PixMo-Cap, despite its descriptive richness, lacks this aspect. Thus, bridging this gap and deriving a "GRIT-like" grounded, hierarchical structure from PixMo-Cap's dense but ungrounded captions defines the preprocessing problem this thesis addresses in Chapter 3.

2.4.1 Concept Distribution & Semantic Analysis of PixMo-Cap

To understand the semantic composition of the dataset, we extracted and analyzed the top noun concepts from the PixMo-Cap captions prior to processing. As shown in Figure 2.4, the vocabulary is heavily dominated by broad contextual descriptors (e.g., *image*, *background*, *scene*) alongside high-frequency spatial indicators (*right*, *corner*, *left*, *center*). This distribution reflects the dense, spatially grounded nature of the caption descriptions compared to standard object-only datasets.

We further visualized the semantic structure of these extracted concepts using a 2D projection of their text embeddings, illustrated in Figure 2.5. The embedding space exhibits clear topological grouping: spatial relational terms (Group 3 and Group 5) form tight clusters corresponding to cardinal locations and layout bounds (*left*, *right*, *corner*, *center*), while image metadata and structural descriptors (Group 1 and Group 4) cluster distinctly in separate semantic regions. This natural separation demonstrates the structured variety present across localized and global semantic domains in the dataset.

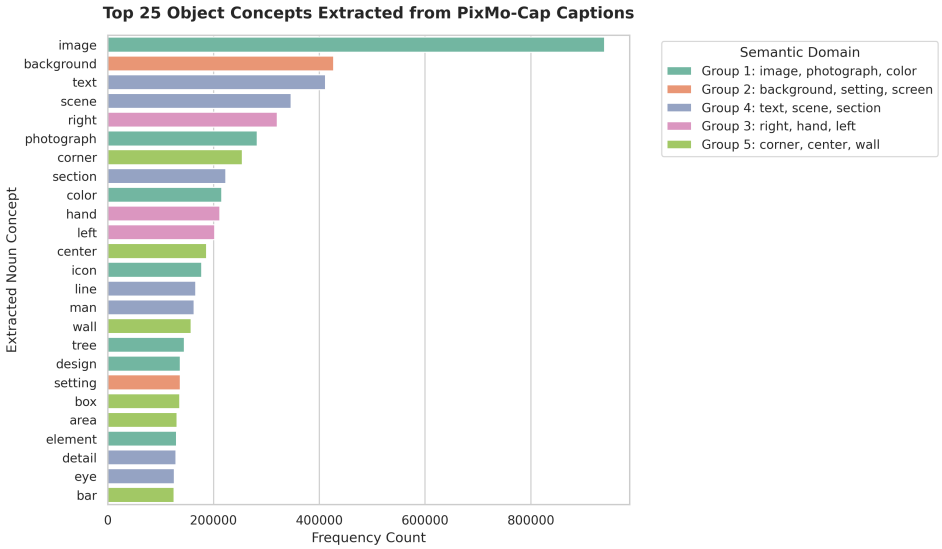


Figure 2.4. Frequency distribution of the top 25 object concepts extracted from PixMo-Cap captions, categorized across five main semantic domains.

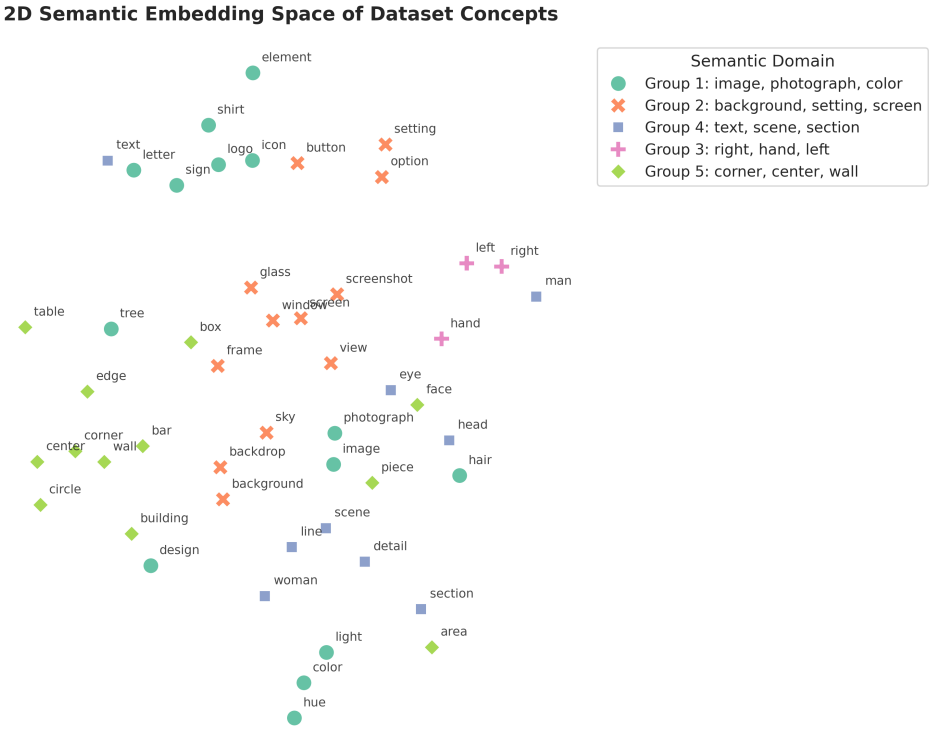


Figure 2.5. 2D projection of the semantic embedding space for dataset concepts, illustrating distinct clustering behavior across the five semantic domain groups.

2.5 State-of-the-Art Dataset Quality Assessment for Vision-Language Data

Because the contribution of this thesis is a preprocessing pipeline rather than a trained model, its evaluation is an exercise of dataset validation, rather than a downstream benchmarking one. This form of evaluation has clear precedent in prior dataset-construction work, which typically combines two complementary approaches: corpus-level statistical characterization, and manual or qualitative inspection of samples.

On the statistical side, dataset papers commonly report summary statistics as evidence of a dataset's scale and structure, often directly against prior work. The GRIT paper [11], for instance, validates its construction pipeline by reporting the number of images, text spans, and bounding boxes it produces, and situates these figures in a direct comparison table against existing grounding datasets such as RefCOCO [12], RefCOCO+ [13], RefCOCOg [14], and Visual Genome [15]. Visual Genome [15] goes further, dedicating an entire section of its paper specifically to "data analysis statistics," separate from both its construction methodology and its downstream benchmarking results. There, region-description statistics (an average of 42 regions per image, each with a bounding box and a descriptive phrase) are reported alongside qualitative observations linking region size to specificity of the description, noting that regions covering a large portion of an image tend to receive more general descriptions, while small regions receive more specific ones. This kind of size-versus-specificity relationship is directly analogous to the box-area and box-count analysis planned for this thesis's own dataset.

Statistical summaries alone can miss systematic failure modes that only become apparent on close inspection. For instance, in an audit of DataComp's CLIP-based filtering pipeline (Section 4), Gadre et al. [23] manually examined image-text samples from the websites that its filter rated most highly, finding that a large share of the top-scoring sources were e-commerce and stock-photo platforms. This is a pattern in the filtered data that statistical pass-rate figures alone did not, and can not reveal, and that additional manual inspection was needed to characterize. Manual, sample-level inspection, particularly of tail or edge cases that a statistical summary would otherwise obscure, is therefore treated in the literature not as a substitute for rigorous evaluation but as a standard complement to it.

Taken together, these previous works establish statistical characterization and manual sample inspection as an established methodology for the validation of a vision-language dataset. Chapter 4, taking those as a guide, applies the combination of box area and count distributions, descriptive histograms, and a hands-on qualitative review of tail samples to assess the output of the preprocessing pipeline developed in Chapter 3.

2.6 Open-Vocabulary Grounding and Vision-Language Architectures

Localizing natural language descriptions within visual scenes requires bridging dense text semantics with spatial region proposals [11, 19]. Traditional object detection frameworks, such as Faster R-CNN [16] or YOLO [17], are restricted to fixed, closed-vocabulary category sets (e.g., COCO's 80 classes [20]) [11, 28]. To support unrestricted natural language queries, modern vision-language models (VLMs) and open-vocabulary grounding detectors reformulate detection as a cross-modal alignment or text-generation problem [18, 19, 26]. These models generally

fall into three architectural paradigms: deep cross-attention networks, autoregressive vision-language sequence models, and decoupled dual-encoder detectors [19, 26, 28].

2.6.1 Deep Cross-Attention Grounding Models

Architectures such as GroundingDINO [19] and MDETR [25] interleave cross-attention operations between image patch tokens and text token embeddings throughout their transformer backbone layers [19]. This early and continuous multi-modal fusion allows phrase tokens and visual features to mutually inform one another across feature depths [19]. Consequently, deep cross-attention models demonstrate strong performance in resolving complex spatial prepositions, attribute bindings, and compositional expressions (e.g., distinguishing "the red mug to the left of the laptop") [19]. However, because image and text tokens are jointly processed at every layer, query text representations cannot be pre-computed independently of the image, introducing significant computational overhead when processing large-scale datasets with diverse text prompts [19].

2.6.2 Autoregressive Vision-Language Models

Unified multi-task VLMs, including Florence-2 [26] and Qwen-VL [27], treat visual grounding as a sequence-to-sequence text generation task [26, 27]. In these models, bounding box coordinates are discretized into spatial location tokens (e.g., `<loc_x1>`, `<loc_y1>`) within the model's vocabulary [26, 27]. Given an image and a textual prompt, the autoregressive decoder outputs bounding box coordinates directly as structured text strings [26, 27]. While this unified paradigm offers high flexibility across down-stream visual tasks (such as region captioning, object detection, and optical character recognition) [26], sequence generation causes token-by-token decoding latency and carries a risk of non-deterministic output formatting during automated batch processing [26, 27].

2.6.3 Decoupled Dual-Encoder Detectors

Decoupled, or dual-encoder, open-vocabulary detectors such as by OWL-ViT and OWLv2 [28] process image and text modalities through independent visual and textual backbone encoders [28]. Cross-modal similarity is evaluated only at the final stage via a lightweight projection head and contrastive similarity score [28]. OWLv2 scales open-world localization through object-level self-training pseudo-labeling (OWL-ST) on web-scale image-text pairs [28]. The primary operational advantage of the dual-encoder architecture is feature decoupling. Text embeddings for candidate phrases can be computed once and cached, allowing open-vocabulary detection over variable text queries to execute in fixed linear time[28].

Each of these three paradigms represents distinct trade-offs between fine-grained compositional grounding accuracy, task flexibility, and computational throughput. In Chapter 3, we evaluate models from each class as candidate components before selecting the final architecture for our offline dataset grounding pipeline [28].

Chapter 3

Methodology

3.1 Introduction of the Pipeline

Constructing a training corpus suitable for hyperbolic vision-language pretraining, specifically HyCoCLIP [10], aside from having the requirement for paired images and captions, needs a predetermined hierarchy for each paired *hierarchical* entity, again in the form of pairs of images and captions. Putting the whole scene together, each text-image pair should be provided with the individual, localized sub-regions and object mentions it contains, related to their originating text spans. PixMo-Cap [22], despite providing long, narration-derived dense captions, does not contain such region-level annotations; therefore, an offline pipeline was needed to derive them. This chapter describes that pipeline with its very final implementation, and, where relevant, the alternative designs that were implemented and evaluated before arriving at it. Two engineering constraints shaped essentially every design decision described below, and are introduced here so that later sections can refer back to them:

- **Throughput over per-sample sophistication:** The pipeline had to process a dataset on the order of hundreds of thousands to low millions of caption-image pairs on a single-GPU, session-limited compute environment (Kaggle, with a hard per-session wall-clock budget and a single T4/P100-class accelerator). Any component of the pipeline whose per-sample cost scaled with the complexity of its output rather than being a fixed-cost forward pass threatened to make the entire pipeline computationally infeasible, regardless of the quality of its output on individual examples.
- **Resumability under interruption:** Given session time limits and the unreliability of fetching several hundred thousand images over the network, no stage of the pipeline could assume it would run to completion in a single, uninterrupted pass. Each stage was therefore designed to persist enough state to disk to resume exactly where it left off, rather than requiring costly reprocessing of already-completed work.

The resulting pipeline itself consists of four stages:

1. Extracting the noun chunks from the raw captions
2. Grounding the extracted noun chunks and referring expressions to image regions using the open-vocabulary object detector OWLv2 [28]

3. Filtering the resulting detections by box size, confidence score, and non-maximum suppression (NMS), and reconciling the surviving boxes with their corresponding text spans

4. Assembling the final training corpus by fetching the source images, cropping the grounded regions, resizing all images to a uniform resolution, and packaging the resulting parent-child image-text pairs into WebDataset shards

The remainder of this chapter discusses each stage in turn (Sections 3.2 through 3.5), including the alternative designs considered at each point and the concrete reasons for their abandonment.

3.2 Stage 1: Noun Chunk Extraction

The first stage processes each raw caption to extract candidate noun chunks: short text spans referring to visually groundable entities, which later serve as open-vocabulary text queries for the grounding stage. This is a purely text-side, CPU-bound stage, and was deliberately kept that way: since every downstream stage depends on its output, any unnecessary GPU or network dependency introduced here would have propagated a bottleneck through the entire pipeline.

3.2.1 Design

Each caption is first split into sentences using a conservative, quote-aware sentence splitter, ensuring that noun chunks and referring expressions cannot later cross a sentence boundary to make sure the long characteristic of PixMo-Cap's narration style is avoided. A dependency parser (spaCy [24], `en_core_web_sm` v3.7) is then applied per sentence to extract noun chunks and, where possible, to expand each one into a fuller referring expression. Expansion proceeds by walking a restricted set of dependency relations outward from the chunk's syntactic root, namely determiners, adjectival and compound modifiers, possessives, and a fixed whitelist of prepositions (*of*, *with*, *wearing*, *holding*, etc.), while explicitly refusing to cross clause boundaries (relative clauses, adverbial clauses, coordination, apposition) or to enter a verb phrase. This distinction matters: a phrase such as "the dog that was barking at the mailman" should not be expanded past "the dog", since the relative clause introduces a separate, non-nominal predication that is not part of the entity being referred to.

3.2.2 Abandoned Alternative: Lexical Blocklist Filtering of Meta-Descriptors

A recurring problem at this stage is meta-descriptive language: text referring to the photograph, the caption, or the image frame itself, rather than to an object actually depicted in the scene (e.g. "the image shows...", "in the top corner of the photo..."). Left unfiltered, such phrases pollute the set of text queries passed to the grounding stage with ungroundable targets.

An earlier version of this stage attempted to solve this problem purely lexically: a fixed list of meta-referential head nouns (*image*, *photo*, *frame*, *corner*, among others) was compiled by manually inspecting a sample of the dataset and noting which nouns recurred as meta-descriptors, and any noun chunk whose head noun matched this list was discarded outright, independent of its syntactic context.

This approach was abandoned because several of these nouns are lexically ambiguous between a meta-referential sense and a genuine, groundable object sense,

and a lexical blocklist has no way of distinguishing the two. The clearest case is *frame*: it is frequently used to describe the boundary of the photograph itself (a meta-descriptor), but can equally refer to an actual object depicted in the scene, such as a picture frame hanging on a wall (a groundable object). Filtering on the head noun alone could not tell these apart, and consistently discarded valid, groundable chunks together with the meta-referential ones it was meant to suppress, reducing the coverage of the resulting grounded dataset.

A second, more structural problem became apparent on further inspection: because the blocklist was compiled from terms observed to cause problems in *this* dataset, it was necessarily incomplete with respect to the broader phenomenon. The noun *border* exhibits the identical ambiguity—an image border versus an actual border visual—yet it was never added to the list, simply because it had not been noticed during manual inspection. A purely lexical, corpus-derived blocklist can only ever cover the specific ambiguous terms it happened to encounter; it does not generalize to the open-ended and continually expanding vocabulary of naturally written captions, and would require indefinite manual maintenance as new ambiguous terms surfaced.

3.2.3 Resolution: Syntactic-Context Filtering

This motivated a shift from lexical identity to syntactic role as the basis for filtering. In the final pipeline, a noun is blocked only when it functions as the *head* of a chunk, i.e when the chunk's syntactic root itself refers to a meta-descriptor, rather than when it merely appears anywhere within a chunk, for instance as a compound or possessive modifier of some other head. Positional framing constructions such as "upper corner" or "left side" are identified separately, via explicit pattern-matching over combinations of framing words (*top*, *left*, *upper*, ...) and framing heads (*corner*, *side*, *portion*, ...), rather than by blocking either word class in isolation. This reduces, though does not entirely eliminate, the ambiguity problem described above: a term such as *frame* used as a genuine head noun ("a picture frame") is still blocked unconditionally under the current implementation, since no additional disambiguating signal (e.g. the presence of material or decorative modifiers) is yet consulted. Extending the filtering logic to consult such local modifiers is left as a direction for future refinement of the extraction stage.

Several other elements were layered onto this syntactic approach. For example, double-quoted spans are treated as atomic units and never split by the parser; apostrophes in possessives are normalized rather than treated as quotation delimiters (*"bird's"* $\rightarrow$ *"birds"*); bare pronoun chunks are dropped entirely; and deictic (*this*, *that*), evaluative (*beautiful*, *stunning*), and commentary (*reminiscent*, *symbolic*) modifiers are stripped from otherwise valid chunks, since these describe the caption author's framing or opinion rather than a visible property of the scene.

3.3 Stage 2: Grounding

The second stage takes the noun chunks and referring expressions produced by Stage 1 and localizes each one to a region of its source image, producing the (image region, text) pairs that HyCoCLIP's region-level entailment objective requires. This stage is by far the most computationally expensive in the pipeline, since it is the only one requiring a GPU forward pass per image, and consequently received the most design iteration.

3.3.1 Engineering Requirements for the Grounding Component

Before comparing candidate architectures, it is worth stating explicitly what was required of this stage, since these requirements directly explain why certain otherwise-capable architectures were unsuitable:

1. **Open-vocabulary querying:** The set of noun chunks varies freely per image and is not drawn from a fixed, closed set of object classes, ruling out closed-set detectors (e.g. Faster R-CNN [16], YOLO [17]) trained on a fixed label set such as COCO's 80 classes [20].

2. **Bounded/fixed-cost inference per image:** Given the throughput constraint stated in Section 3.1, the cost of grounding a single image needed to scale, at worst, linearly with the number of candidate phrases for that image, and ideally sublinearly if phrase representations could be reused or cached.

3. **Deterministic and structured output:** The output of this stage needed to be a small, fixed-format set of (box, score, phrase index) tuples suitable for direct programmatic filtering (Stage 3), rather than free-form natural language requiring downstream parsing.

3.3.2 Candidate Architecture 1: Deep Cross-Attention Grounding (GroundingDINO)

GroundingDINO [19] and related deep cross-modal grounding models (e.g. MDETR [25]) interleave cross-attention between image patches and text tokens throughout the network backbone, rather than processing the two modalities independently. This early-fusion design allows text and image representations to inform one another at every layer, which typically yields stronger performance on compositional or spatially qualified referring expressions, for instance, correctly resolving "the red mug to the left of the laptop" by jointly reasoning over color, object identity, and spatial relation, rather than treating each attribute independently.

This architecture was implemented first, but was abandoned for this pipeline primarily on efficiency grounds. Because text and image tokens are fused jointly at every layer, text representations cannot be pre-computed and cached independently of the image being processed: every image requires a full re-processing of its associated text queries through the shared cross-attention stack. At the scale of this dataset this joint-processing requirement produced GPU memory and runtime costs that made full-dataset processing impractical within the available compute budget, independent of the (generally strong) quality of the individual detections it produced on smaller samples.

3.3.3 Candidate Architecture 2: Autoregressive Vision-Language Grounding (Florence-2, Qwen-VL)

A second family of candidates consisted of autoregressive vision-language models capable of producing bounding-box coordinates directly as generated text tokens, such as Florence-2 [26] and Qwen-VL [27]. In principle, these models are attractive for this task because a single model can both interpret an open-ended natural-language query and emit a structured, textual representation of the resulting box, without requiring a separate detector architecture.

In practice, this family of approaches was abandoned for two related reasons, both stemming from the autoregressive nature of the output. First, because bounding-box coordinates are generated token-by-token rather than predicted in a single

forward pass, per-image inference latency was substantially higher than either the deep cross-attention or dual-encoder alternatives, again violating the fixed-cost-per-image requirement stated above. Second, and more consequentially for data quality, autoregressive box generation is not *formatting*-guaranteed: outputs occasionally failed to conform to the expected coordinate format, requiring additional parsing and validation logic, and silently dropping or corrupting a non-trivial fraction of samples at scale. Between the latency cost and the parsing overhead, neither Florence-2 nor Qwen-VL used purely as detectors were judged suitable for full-dataset processing.

3.3.4 Candidate Architecture 3: GroundingDINO Detection with VLM Re-Captioning

A third variant combined GroundingDINO [19] for region detection with Qwen-VL [27] used not as a detector but as a captioner: rather than reusing the original caption's text span for a detected region, Qwen-VL was prompted to synthesize a fresh, self-contained caption describing each cropped ("child") image region, with the intention of producing cleaner, more descriptive region-level text than a raw substring of the original dense caption could offer. This produced qualitatively reasonable captions when evaluated on small samples.

This approach was nonetheless abandoned, again on scalability grounds, and for a subtly different reason than the previous two candidates: whereas GroundingDINO alone and Florence-2/Qwen-VL-as-detector incur a cost proportional to the number of *images*, this variant additionally required one VLM captioning call per *detected region*. Since a single image frequently yields several detected regions, this multiplied the number of required VLM forward passes by the average number of detections per image, rather than processing each image once regardless of how many regions it contained. This shift from a per-image to a per-region cost structure made the approach computationally prohibitive at the scale required for this thesis, given the available compute budget and session time limits described in Section 3.1.

3.3.5 Selected Architecture: OWLv2 (Dual-Encoder)

The architecture ultimately adopted, OWLv2 [28], belongs to a third family of vision-language architectures: *dual-encoder*, or decoupled, models, in which an image encoder and a text encoder process their respective modalities independently and are combined only at a late stage, via a lightweight projection and similarity score, rather than through interleaved cross-attention. This design directly satisfies the fixed-cost-per-image requirement above: text embeddings for the noun chunks associated with a batch of phrases can be computed once, independently of the image, and the (comparatively expensive) image encoder forward pass can then be run without incurring any further per-phrase text-processing cost during that pass. This decoupling is also what enables OWLv2's pretraining recipe (OWL-ST [28]) to scale to over a billion unannotated web image-text pairs using pseudo-labels, rather than requiring the explicit phrase-region aligned supervision that deep cross-modal architectures typically depend on.

This efficiency is not free: because the two modalities never directly attend to one another during inference, dual-encoder architectures are comparatively weaker than deep cross-attention architectures at resolving compositional or spatially qualified queries (e.g. distinguishing "a red car and a blue truck" from "a blue car and a red truck"), a pattern observed on benchmark datasets such as RefCOCO [12] and RefCOCOg [14] that specifically test this kind of compositional grounding. For this pipeline, however, the text queries derived from Stage 1 are short, single-entity noun

chunks rather than complex, multi-attribute referring expressions, which substantially reduces the practical impact of this limitation: the queries this pipeline needed to ground rarely required the fine-grained compositional reasoning that motivates deep cross-attention architectures in the first place. Combined with the throughput and pretraining-scalability advantages described above, this made OWLv2 [28] the architecture best matched to the specific requirements of this pipeline.

3.4 Stage 3: Detection Filtering

Open-vocabulary object detectors such as OWLv2 [28] are often operated at lower confidence thresholds to maximize recall, producing detections that must be filtered before the data is usable for training. Three filters are applied in sequence, each addressing a distinct failure mode.

First, a minimum box-side filter discards detections whose width or height, in pixels, falls below a fixed threshold, removing degenerate, near-point detections that carry no meaningful visual content once cropped. Second, a confidence threshold discards detections whose score falls below a cutoff. This threshold was deliberately calibrated empirically against the score-percentile distribution observed on this dataset with OWLv2, rather than reusing a threshold reported in prior work for a different detector (e.g. the 0.65 threshold commonly used for GLIP [18]): detection score distributions are not directly comparable across model architectures and training regimes, so a threshold tuned for one detector does not transfer reliably to another. Third, class-agnostic non-maximum suppression (NMS) [29] is applied across all surviving detections within an image, removing duplicate or heavily overlapping boxes that arise when the same underlying region is independently matched to several near-synonymous query phrases (e.g. “the dog” and “the brown dog” both matching the same region).

Records for which no detections survive this three-stage filter are dropped entirely rather than retained with an empty region set, since HyCoCLIP’s region-level objective requires at least one valid region per training sample [10]. The surviving detections are reconciled back to their originating noun chunk or referring expression via the phrase index returned by OWLv2, ensuring that every retained box remains paired with the exact text span used to localize it, preserving the provenance needed for the region-caption pairing constructed in Stage 4.

3.5 Stage 4: Dataset Assembly

The final stage converts the filtered (region, text) pairs from Stage 3 into the training-ready WebDataset format [30] consumed by the HyCoCLIP training pipeline, and is where the resumability requirement from Section 3.1 is most directly engineered into the implementation, since this stage is dominated by network I/O (re-fetching several hundred thousand images) rather than GPU computation.

For each surviving record, the source image is re-fetched over the network, and each retained bounding box is cropped from the original, full-resolution image in order to preserve the maximum available visual detail in each region crop. Both the full ("child") image and each cropped ("parent") region are then independently resized via a short-edge resize followed by a center crop to a fixed 224×224 resolution, matching the input preprocessing pipeline expected by ViT-based vision encoders [1, 10]. The resulting child image, its original caption, and its associated set of parent crops with their matched text spans are serialized into WebDataset-format [30] `.tar` shards, each holding a fixed number of samples.

Two engineering decisions in this stage were made specifically to satisfy the resumability requirement under the session-limited compute environment used for this thesis. First, shard writing persists its progress (current line index, current global sample index, and current shard index) to disk after every completed shard, so that an interrupted run can resume processing from the exact sample at which it stopped, rather than either restarting from the beginning of the input or requiring manual bookkeeping between sessions. Second, image fetching and per-region processing are parallelized across a bounded pool of worker threads with a fixed maximum number of in-flight requests, trading a small amount of memory overhead for substantially higher network throughput than a strictly sequential fetch-crop-resize loop would achieve, without allowing an unbounded number of concurrent network requests to overwhelm the available bandwidth or memory.

3.6 Summary

Across all four stages, two engineering principles recur and were treated as first-class design constraints rather than incidental implementation details: *scalability of per-sample cost* (favoring architectures and algorithms whose cost per image or per caption remains bounded and, where possible, cacheable, over those whose cost scales with the number of detected regions or the complexity of the generated output) and *robustness to interruption* (favoring designs that persist enough state to resume exactly where they left off, over designs that assume uninterrupted, single-session execution). Both principles directly reflect the constraints of the environment in which this pipeline was developed, and both recur explicitly in the architecture and alternative-design discussion in Sections 3.2 through 3.5 above.

Chapter 4

Statistical Analysis of the Resulting Dataset

The pipeline described in Chapter 3 turns PixMo-Cap's dense, ungrounded narrations into a set of images, captions, and region-phrase pairs. Whether this output is actually suitable for training HyCoCLIP cannot be read off from the design of the pipeline alone, and must be checked on the data that the pipeline produced. This chapter carries out that check. Since the contribution of this thesis is a dataset construction pipeline rather than a trained model, the evaluation takes the form of a dataset validation, following the two-part methodology reviewed in Section 2.5: a statistical characterization of the corpus, and a manual inspection of individual samples.

4.1 Evaluation Rationale

HyCoCLIP expects every training sample to consist of a full image, its caption, and at least one pair made of an image box and a text box, where the box is a genuinely more specific part of the image, and the text box a genuinely more specific part of the caption (Section 2.2). A dataset satisfies this requirement only if four conditions hold at the same time: it is large enough to be useful, the regions are well-formed geometric objects, the phrases are well-formed textual objects, and the two match. Each of these conditions translates into a question that guides the analyses of this chapter:

Q1 **Scale:** How many samples, boxes, and phrases does the final dataset contain, and how does this compare with GRIT?

Q2 **Structure:** How many boxes does a sample carry, and how does this compare with the dataset HyCoCLIP was originally trained on?

Q3 **Text quality:** Are the text boxes short and specific enough to be more specific than the caption they come from, yet informative enough to be useful?

Q4 **Geometry:** Do the boxes occupy a sensible portion of the image, that is, neither the whole image nor a few pixels?

Q5 **Confidence:** Is the confidence threshold calibrated in Stage 3 placed reasonably, and do the scores carry information about quality?

Analysis	Question	Design decision examined
Dataset scale	Q1	Overall output of the four-stage pipeline, compared with the reference dataset used for HyCoCLIP
Boxes per sample	Q2	Requirement of at least one region per sample; NMS; effect of the cap of 30 candidate phrases per image
Phrase length	Q3	Dependency-tree expansion, clause-boundary truncation, single-sentence isolation
Box area ratio	Q4	Minimum box-side filter; cropping at full resolution and resizing to 224×224
Confidence histogram	Q5	Empirically calibrated confidence threshold for OWLv2
Manual review of samples	Q6	Meta-descriptor filtering, modifier stripping, choice of OWLv2 over deep cross-attention grounding

Table 4.1. Overview of the analyses of this chapter, the question each one answers, and the design decision of Chapter 3 it tests.

Q6 Correctness: When examined directly, are the region-phrase pairs actually right, and do the filters of Stage 1 remove the text-side noise as they were designed for?

Table 4.1 links each analysis to the question it answers and to the design decision of Chapter 3 that it puts to the test. The analyses were, more than convenience, chosen so that each of them examines a specific choice made in the pipeline, so that a disappointing result points back to a concrete component that can be revised.

4.2 Experimental Setup

4.2.1 Unit of Analysis and Data Access

The unit of analysis is a *sample*: a single PixMo-Cap image paired with its caption and retained region-phrase pairs. With the exception of the box area ratio, all corpus statistics are computed directly from the per-sample metadata persisted alongside the WebDataset shards in Stage 4 (including bounding box coordinates in the original frame, image dimensions, text spans, and detection scores). Because this pass operates strictly on structured metadata without decoding image pixels, its computational overhead is negligible compared to the pipeline itself.

The box area ratio is computed instead from the intermediate outputs of Stage 3. Because Stage 4 crops and standardizes all visual assets to a fixed 224×224 canvas, calculating spatial coverage post-Stage 4 would distort the original relative area of bounding boxes. Computing this metric from Stage 3 metadata directly aligns with the throughput constraints established in Section 3.1: Stage 3 outputs lightweight coordinate metadata prior to heavy image cropping and TAR archive serialization. Evaluating spatial coverage at this stage requires only an I/O-light metadata pass and allows quality statistics to be re-evaluated instantaneously whenever upstream filtering parameters ($\theta_{\text{conf}}, \theta_{\text{NMS}}$) are tuned, without requiring a full re-execution of Stage 4 sharding.

Dataset	Images	Boxes	Text spans	Boxes / image
GRIT (full) [11]	91M	>137M	115M	≈1.5
PixMo-Cap grounded (ours)	608,290	4,397,937	4,397,937	7.23

Table 4.2. Scale of the resulting dataset, counted from the final WebDataset shards, compared with GRIT.

4.2.2 Reporting Conventions

The quantities studied in this chapter (boxes per sample, phrase length, area ratio, confidence score) are all bounded below and typically right-skewed. For such quantities the arithmetic mean alone is a misleading summary, since a small number of extreme samples can shift it considerably. Every distribution is therefore reported with its mean, median, standard deviation, and the 5th, 25th, 75th and 95th percentiles, and is always accompanied by a histogram. The bin width of each histogram follows the Freedman–Diaconis rule [31],

$$h = 2\,\frac{\mathrm{IQR}(x)}{n^{1/3}}, \tag{4.1}$$

where IQR is the interquartile range and n the number of observations. This rule depends on the spread of the data through a robust statistic instead of the standard deviation, so that it is not inflated by the very tails that are of most interest here. Where a distribution spans several orders of magnitude, as the area ratio does, the horizontal axis is additionally shown on a logarithmic scale. Relationships between two variables are quantified with Spearman's rank correlation coefficient ρ [32], which measures monotonic association without assuming linearity or normality and is therefore appropriate for skewed data.

4.3 Dataset Scale

The first analysis addresses Q1 and follows the practice of dataset papers of reporting the scale of the resulting corpus (Section 2.5). Table 4.2 places the dataset next to GRIT, using the same three quantities (images, bounding boxes, text spans) with which the GRIT paper compares itself against earlier grounding datasets [11]. GRIT contains about 91 million images with more than 137 million boxes, i.e. roughly 1.5 boxes per image [11]. The dataset of this thesis is several orders of magnitude smaller, which is a consequence of the compute budget described in Section 3.1 and of the size of PixMo-Cap itself, and should not be read as a quality statement. The figures that are comparable across the two datasets are therefore the *per-sample* ones, analyzed in the following sections, rather than the absolute counts. For context, Visual Genome, whose regions are annotated manually, contains an average of 42 region descriptions per image [15], which represents a natural upper bound for what a fully automatic pipeline can be expected to approach.

Since the analysis starts from the finished shards, it characterizes what the pipeline *delivers* and not how much material each stage removed along the way. The consequences of this choice are discussed in Section 4.10. The only retention figure that can be given is the overall one: of the 712,000 images of PixMo-Cap [2], 608,290 (85.43%) are part of the final dataset. The remaining 103,710 images (14.57%) are not lost at a single point. A record can leave the dataset in four places: Stages 1 and 3 act as deliberate precision filters, while Stages 2 and 4 lose records through

image retrieval. In Stage 1, a caption is discarded when none of its noun chunks yields a valid referring expression (Section 3.2); given the average caption length of about 196 words [22], this should be rare. PixMo-Cap is distributed as image URLs that must be downloaded separately [2], so in Stages 2 and 4 an image is skipped if its host is unreachable, the request times out, or the file is smaller than 32 pixels on its shorter side. For Stage 4 the loss can be quantified: the boxes in the output of Stage 3 sum to 4,427,459, against 4,397,937 in the final shards (0.67% fewer), so this step accounts for less than one percentage point.

The largest share is most likely due to Stage 3. Since the raw detection scores are low (mean 0.1976), the threshold of 0.30 removes most detections, and an image without any surviving box cannot be kept, because HyCoCLIP requires at least one region per sample (Section 3.2.3). Which cause dominates cannot be determined, as the number of records after each stage was not recorded (Section 4.10). The retention of 85.43% should therefore be read as the combined effect of two precision filters and of the availability of the source images, not as a measure of how many captions are inherently ungroundable.

4.4 Boxes per Sample

4.4.1 Motivation

The number of boxes per sample decides how much region-level supervision each image contributes. HyCoCLIP's objective is defined over an image box and a text box per sample, and it constrains the pairs (I, T), $(I^{\text{box}}, T^{\text{box}})$, (I, I^{box}) and (T, T^{box}) [10]. A sample with a single box provides exactly the required structure and nothing more. A sample with several boxes provides a *pool* from which different boxes can be drawn, so that the same image can contribute varied compositional signal without being repeated. Too few boxes on average therefore weakens the benefit of the whole grounding effort, whereas a very high count usually indicates redundant, heavily overlapping boxes that survive NMS with different phrases, or a systematic lack of selectivity in the filter. Both extremes are visible only in the full distribution, not in the mean.

This analysis also gives a first indication of the effect of two pipeline decisions. The cap of 30 candidate phrases per image imposes a hard ceiling on the count, and a spike near this ceiling would show that the cap and not the image content is limiting the result. The maximum observed number of boxes and the share of samples close to the ceiling are therefore reported. Conversely, PixMo-Cap's captions average around 196 words [22], i.e. far longer than the web captions from which GRIT was built, so more candidate phrases, and hence more boxes per image, are expected than for GRIT.

4.4.2 Comparison with GRIT and Contrastive Baselines

In a standard CLIP-style contrastive corpus [1] every sample is an image paired with one caption, with no sub-image or sub-caption structure. The number of boxes per sample is therefore zero by definition, and the hierarchical terms of HyCoCLIP's loss have nothing to operate on. This comparison is degenerate, and is stated here only to define the baseline that the grounding pipeline improves upon. The meaningful comparison is with GRIT [11], the dataset from which HyCoCLIP obtained its hierarchy. For GRIT, the average of about 1.5 boxes per image derived from the published totals serves as a first reference, and the full distribution is obtained from the sample described in Section 4.2.

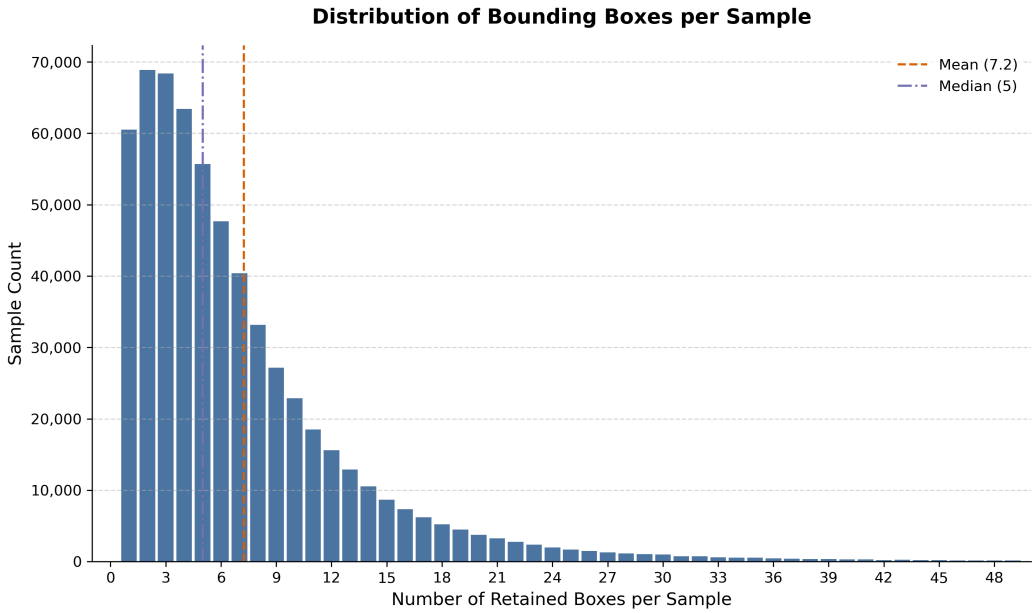


Figure 4.1. Distribution of the number of boxes per sample in the resulting dataset.

4.4.3 Results

Figure 4.1 shows the distribution of the number of parent crops per sample. Across the 608,290 analyzed samples, a sample carries an average of 7.23 parent crops (median 5.00, ranging from a minimum of 1 to a maximum of 369, with a standard deviation of 7.40). The distribution is highly right-skewed, reflecting the extensive, fine-grained details inherent in PixMo-Cap's captions. Additionally, the presence of a maximum value of 369 parent crops indicates that dense imagery can dramatically exceed the average.

4.5 Length of the Text Boxes

4.5.1 Motivation

The text box of a sample has to be a more specific description than the full caption that contains it, since HyCoCLIP treats a phrase as entailed by the sentence it was taken from [10]. The length of the phrases is a direct, measurable proxy for this property and for the behavior of the Stage 1 expansion. Phrases that are too short (typically a bare noun, such as "the dog") carry little information and are easily matched with many regions, while phrases that are too long approach the size of a whole sentence, blur the boundary between the text box and the full caption, and are more likely to contain material that does not describe the object at all. The dependency-based expansion with a preposition whitelist, the refusal to cross clause boundaries, and the single-sentence isolation (Section 3.2) were designed precisely to keep phrases between these two extremes. The histogram of phrase lengths shows whether they succeed.

There is also a practical reason for interest in the length. Standard CLIP-style text encoders [1] process a fixed maximum context window (77 tokens). PixMo-Cap's full captions, at an average of about 196 words [22], exceed this limit, so the model

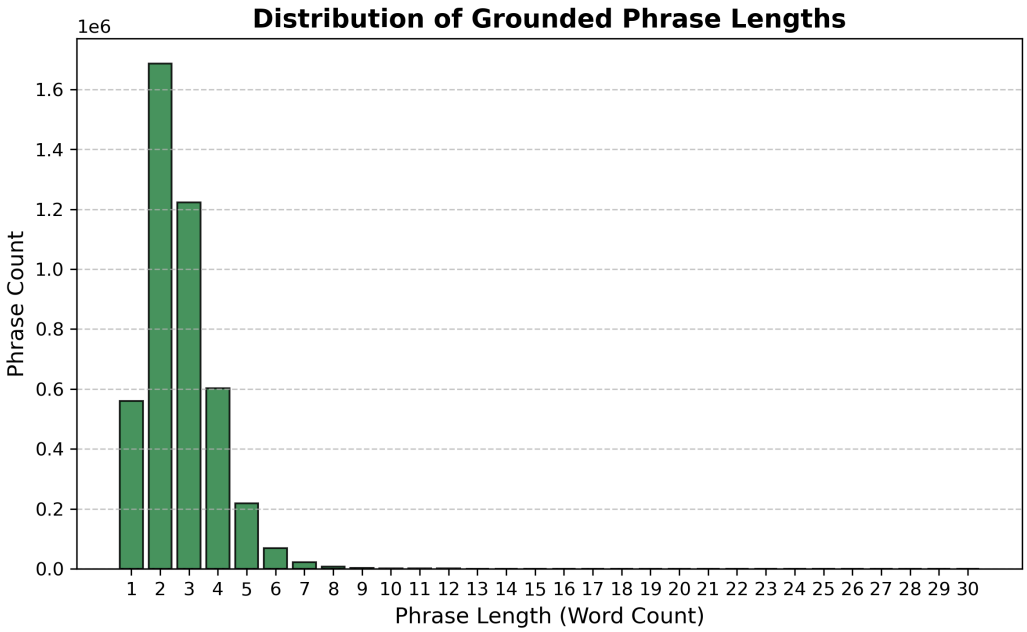


Figure 4.2. Distribution of the length of the text boxes, in words.

can only see part of them. The referring expressions, on the other hand, are expected to fit in this window entirely, which is an additional benefit of region-level text that is not obtained from the full captions.

4.5.2 Results

Figure 4.2 shows the histogram of the length of the text boxes in words, with all boxes of the dataset counted individually (that is, for each box, not for each caption). The median length is 3.00 words (mean 3.22, 95th percentile 7.00), against an average full caption length of about 196 words [22], so the text box is on average only 1.64% as long as its caption. 12.69% of the phrases consist of a single word and 5.00% exceed 7 words.

4.5.3 Vocabulary of the Phrases

Length alone does not say whether the phrases are diverse or whether the filters have left systematic artifacts. For this reason, the head nouns of the phrases are additionally counted. Table 4.3 lists the 20 most frequent heads. This check is aimed at the residual noise of Stage 1: the syntactic filter blocks meta-referential heads, but a term such as *frame* is still blocked unconditionally, while others such as *border* were not on the original list (Section 3.2.2). A head that appears prominently among the top ranks and that is not a concrete, groundable object would show that a filter is missing. Conversely, the absence of terms such as *picture frame* shows the recall cost of the acknowledged limitation of the head-noun blocking.

Rank	Head noun	Count	% of boxes
1	windows	45,517	1.13%
2	text	44,596	1.10%
3	trees	34,995	0.87%
4	lights	33,361	0.83%
5	background	26,478	0.66%
6	letters	25,386	0.63%
7	bar	23,722	0.59%
8	books	22,902	0.57%
9	icon	21,913	0.54%
10	area	21,241	0.53%
11	clouds	20,894	0.52%
12	leaves	20,658	0.51%
13	people	20,509	0.51%
14	rocks	19,593	0.48%
15	eyes	17,905	0.44%
16	hair	17,403	0.43%
17	hand	16,416	0.41%
18	wall	16,090	0.40%
19	floor	15,507	0.38%
20	surface	15,459	0.38%

Table 4.3. The 20 most frequent head nouns among the retained text boxes. Together they cover 11.89% of all boxes.

4.6 Box Area Ratio

4.6.1 Motivation and Definition

The box area ratio measures how much of the image a box covers. For a box with width w and height h in an image of width W and height H, all in pixels of the *original* image,

$$r = \frac{w \cdot h}{W \cdot H} \in (0, 1]. \tag{4.2}$$

The ratio is computed on the original image and not on the 224×224 version because Stage 4 crops from the full-resolution image before any resizing (Section 3.5), so the original frame is the one that determines what the crop actually contains.

The distribution of r tests two opposite failure modes. At the upper end, a box with r close to 1 covers almost the whole image, so the image box is barely more specific than the image and the entailment relation (I, I^{box}) degenerates. This can occur when a phrase describes something that fills the frame or when the detector returns a large box for a generic query. At the lower end, very small boxes contain little visual information once cropped, and are, when resized to 224×224, upsampled by a large factor. For a crop with shorter side s pixels, the short-edge resize scales by $224/s$, so the crop is enlarged by that factor; when it is far above one, the input to the vision encoder is blurry and does not resemble the images it was trained on. The minimum box-side filter of Stage 3 was introduced to remove the extreme cases of this kind, and this analysis measures how effective it is. As a further consequence of the same preprocessing, the short-edge resize followed by a center crop discards part of an elongated box: for a box with aspect ratio $a = \max(w, h)/\min(w, h)$, only a fraction $1/a$ of the box is left in the final crop. The distribution of a is therefore reported as well.

A second reason for studying the area ratio comes from Visual Genome, whose

authors observe that regions covering a large part of the image tend to be described in more general terms and small regions in more specific ones [15]. A dataset in which this relationship also holds is consistent with the hierarchical structure that HyCoCLIP relies on; this is checked in Section 4.8.

4.6.2 Results

Figure 4.3 shows the distribution of the area ratio r on a linear and on a logarithmic axis. Across the 4,397,937 analyzed boxes, the median area ratio is 0.0380 after clipping coordinates to [0,1] (mean 0.1540, standard deviation 0.2524, with values ranging from a minimum of -0.0907 due to coordinate formatting to a maximum of 0.9966). The clipping step eliminates raw coordinate artifacts and ensures valid bounding box area calculations across all evaluation samples. A share of 1.33% of the boxes covers more than 90% of the image, and 50.15% covers less than 1%. These two thresholds are used only for reporting and are not additional filters. The histogram reveals a distribution heavily concentrated on very small, fine-grained objects, with half the dataset covering less than 1% of the image area. Conversely, the near-full-image mass is a lot smaller, confirming that degenerate bounding boxes are rare. The median upsampling factor of the crops is 5.13, with 81.15% of the boxes enlarged by a factor of more than two. Because standard vision encoders (such as ViT in CLIP) downsample images to fixed grid resolutions, these localized regions must be significantly magnified to retain fine visual details, indicating that the training framework must effectively handle extreme scale variance and bridge tiny object patches with global context. Finally, the aspect ratio a has a median of 1.63, and for 37.65% of the boxes it is above two. This demonstrates that many extracted regions correspond to narrow, elongated, or wide framing structures (e.g., limbs, long objects, or panoramic sweeps) rather than neat square crops; consequently, preprocessing pipelines like center-cropping or padding cut across major portions of these regions, requiring models to possess robust spatial awareness to handle non-standard bounding box geometries without distorting the underlying relational semantics.

4.7 Detection Confidence

4.7.1 Motivation

Stage 3 discards detections below a confidence threshold that was calibrated empirically on the percentile distribution of OWLv2 scores in this dataset, and not taken over from a threshold reported for a different detector such as GLIP [18] (Section 3.4). That choice was justified by the fact that score distributions are not comparable across detectors, but the resulting histogram was never inspected as a whole. Two questions remain open. First, where does the mass of the retained detections lie relative to the cutoff? If most retained boxes sit just above the threshold, the dataset is dominated by borderline detections, and the exact value of the cutoff matters a lot. Second, do the scores mean anything about the quality of a box? The second question is not trivial, because the scores of neural networks are in general not calibrated probabilities [33], and this is even more so for an open-vocabulary detector applied to phrases outside its training distribution. The scores are therefore treated in this thesis as an ordinal indicator, not as the probability of being correct, and this interpretation is tested in the manual review by comparing the precision of samples from different score ranges.

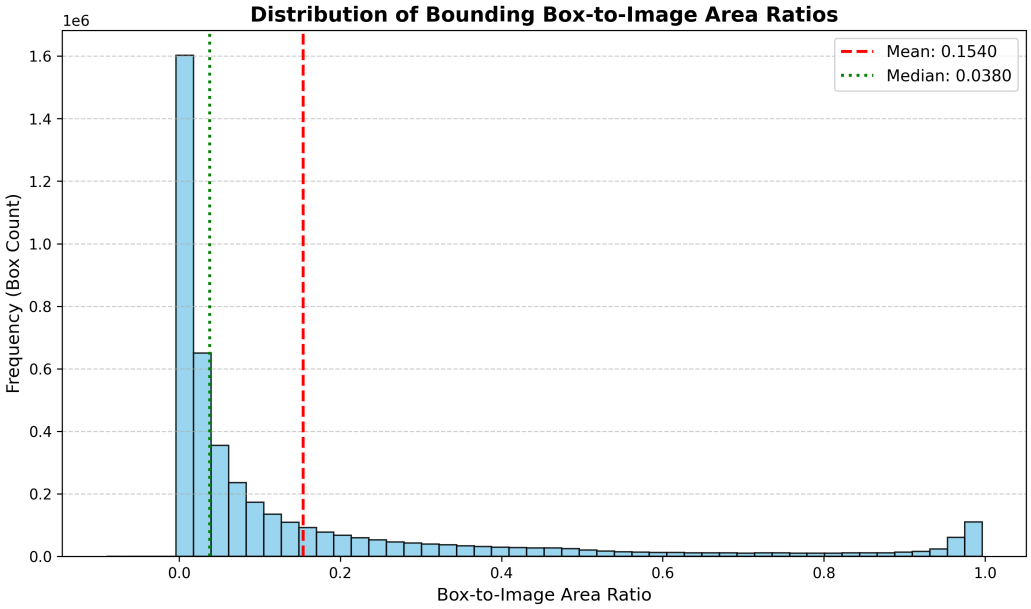


Figure 4.3. Distribution of the box area ratio with respect to the original image.

4.7.2 Results

Figure 4.4 and Figure 4.5 illustrate the distribution of detection confidence scores across the dataset before and after applying the filtering threshold. As shown in Figure 4.4, detection confidence scores across the raw dataset are generally low overall, yielding an initial mean score of 0.1976. Crucially, the tail of the distribution drops off sharply past 0.6, indicating that enforcing a conventional high-confidence threshold (such as 0.6 or higher) would result in a severe loss of region annotations. To preserve the dense, fine-grained spatial coverage necessary for training while filtering out low-quality noise, we empirically set the retention threshold to 0.30. As depicted in Figure 4.5, truncating the distribution at 0.30 eliminates the massive concentration of sub-0.30 noise while retaining a substantial volume of candidate regions, elevating the mean confidence of the retained dataset to 0.4262. The heavy concentration of boxes immediately following the 0.30 threshold demonstrates that the dataset relies significantly on these moderately confident detections to maintain high region density.

4.8 Relationships Between the Variables

To examine the monotonic relationships between spatial, confidence, and linguistic properties of the extracted bounding boxes, pairwise Spearman rank correlation coefficients (ρ) [32] were calculated across the dataset of retained region proposals ($c \geq 0.30$). Spearman's rank correlation was chosen over Pearson's linear correlation because spatial metrics (area ratios) and text metrics (phrase lengths) exhibit heavy right-skewness and non-linear distributions, making non-parametric rank evaluation necessary. Table 4.4 summarizes the empirical rank correlation coefficients alongside their initially hypothesized signs.

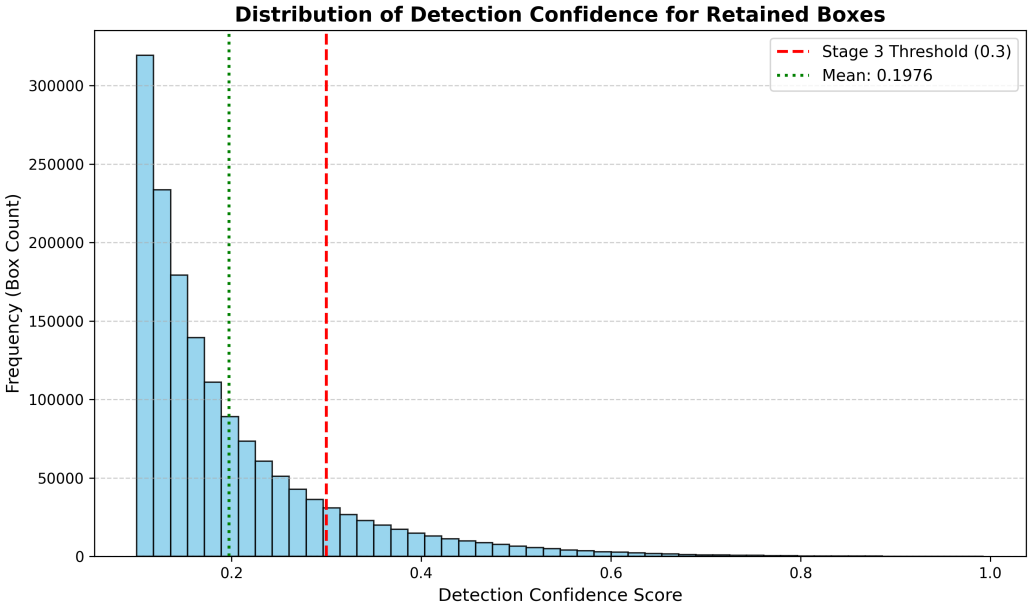


Figure 4.4. Distribution of the detection confidence of the retained boxes before filtering.

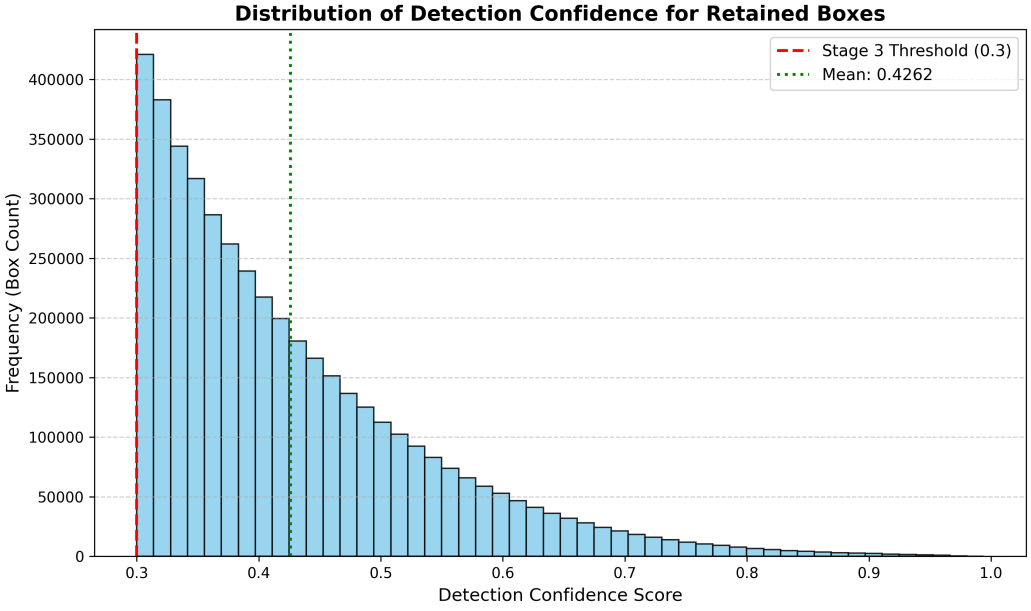


Figure 4.5. Distribution of the detection confidence of the retained boxes after applying empirical filtering threshold of 0.3. The vertical line marks the threshold calibrated in Stage 3.

Table 4.4. Rank correlation between pairs of box-level variables.

Pair of variables	**Expected sign**	**Spearman ρ**
Phrase length vs. Area ratio	Negative	0.1356
Confidence vs. Phrase length	Negative	0.0062
Confidence vs. Area ratio	Positive	0.2015

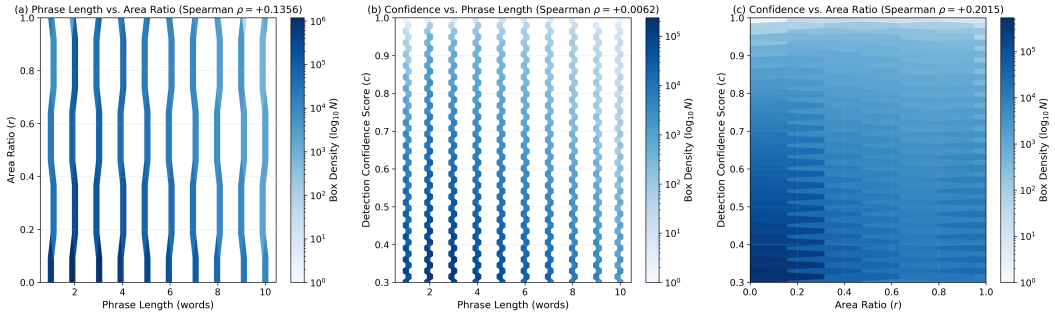


Figure 4.6. Joint distributions of box-level variables. Hexagonal binning is used because the number of boxes is too large for a scatter plot to be readable.

4.8.1 Detection Certainty and Spatial Scale

Detection confidence correlates weakly and positively with the box area ratio ($\rho = 0.2015$). A possible explanation is that larger regions give the detector more visual context, whereas very small regions are harder to localize. Since detector scores are not calibrated probabilities [33], the correlation describes a trend in the ranking of scores rather than in accuracy.

4.8.2 Orthogonality of Detection Confidence and Textual Complexity

The rank correlation between detection confidence c and phrase length L yields a coefficient of $\rho = +0.0062$, indicating that these two variables are statistically independent. The confidence score of the object detector operates entirely independently from the linguistic length of its text annotation. Whether a region is annotated with a concise single-word noun (e.g., *"wheel"*) or a complex descriptive noun phrase (e.g., *"the gray steering wheel with visible controls"*), the detector's ability to localize the object depends strictly on visual clarity rather than the semantic depth or character count of the paired caption.

4.8.3 Revisiting the Scale–Text Complexity Hypothesis

The relationship between phrase length L and area ratio r produced a weak positive correlation ($\rho = +0.1356$), diverging from the initial theoretical hypothesis of a negative relationship. The original assumption posited that smaller, highly localized crops would require longer, more detailed descriptions to disambiguate fine-grained micro-objects from surrounding clutter. However, the empirical data reveals a structural pattern that is weakly consistent with a textual hierarchy:

1. **Atomic Entity Labeling in Micro-Crops:** Very small bounding boxes ($r < 0.01$) dominate when it comes to capturing isolated, single-concept entities (e.g., *"eye"*, *"button"*, *"logo"*). These fine-grained regions are naturally described using brief, 1- to 2-word atomic nouns.

2. **Compositional Richness in Macro-Crops:** Larger bounding boxes ($r > 0.15$) encompass multi-object interactions, visual contexts, and spatial relationships (e.g., *"a man wearing a purple shirt sitting near the window"*). The larger spatial footprint provides the visual real estate required to justify multi-word relational descriptions.

4.8.4 Implications for HyCoCLIP Pretraining

These correlation patterns have direct structural implications for multi-scale vision-language pretraining in HyCoCLIP [10]:

- **Scale Variance Resiliency:** Because small regions ($\rho = +0.2015$) suffer from lower detection confidence scores, pretraining loss formulations must prevent high-confidence large regions from dominating gradient updates over lower-confidence fine-grained details.
- **Hierarchical Text-Image Alignment:** The positive alignment between area ratio and phrase length ($\rho = +0.1356$) indicates that PixMo-Cap weakly exhibits a natural textual hierarchy, suggesting a subtle rather than definitive relationship ($R^2 < 0.02$). Larger crops provide longer, compositionally dense grounding targets suitable for global contrastive alignment, whereas fine-grained crops serve as short, highly localized concept anchors suitable for regional representation learning.

4.9 Manual Review of Samples

Statistics describe how the boxes and phrases are distributed, but they cannot say whether a given box actually contains what its phrase describes. Section 2.5 also showed that statistical summaries can conceal systematic failures that appear only on close inspection, as in the audit by Gadre et al. [23] of the DataComp filtering pipeline. For this reason the statistical analysis is complemented by a manual review, whose design is described here.

4.9.1 Sampling Design

The manual review employs a stratified design partitioning a total evaluation budget of $N = 180$ items into two complementary sample types: an unbiased *simple random sample* baseline and eight targeted *tail strata*.

For proportion estimation, sample sizes are standardly guided by Cochran's formula [34],

$$n = \frac{z^2\, p(1-p)}{e^2}, \tag{4.3}$$

with $z = 1.96$ for a 95% confidence level and the conservative assumption of maximum variance $p = 0.5$. While an unstratified random sample of $n = 180$ would yield a theoretical margin of error of $e = \pm 0.073$, allocating the full review budget solely to random items under-represents rare pipeline failure modes. All reported proportions in this review are computed using Wilson score confidence intervals [35],

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \tag{4.4}$$

which provide superior coverage properties over standard normal approximations when evaluating small sample sizes ($n_s = 20$) or proportions near zero or one.

The second component consists of eight *tail strata*. Because a pure random sample captures few extreme cases in which model failures are most concentrated, samples of $n_s = 20$ boxes are drawn from each stratum listed in Table 4.5, alongside a random baseline stratum of $n_{\text{rand}} = 20$, yielding the combined $N = 180$ review

Stratum	Selection rule	Failure mode probed
Random	Uniform over all boxes	Overall precision
Lowest confidence	Lowest decile of scores	Are borderline detections acceptable, i.e. is the threshold well placed?
Highest confidence	Highest decile of scores	Are high scores systematically wrong (e.g. for whole-image boxes)?
Largest boxes	$r > 0.90$	Degenerate image-box relation
Smallest boxes	$r < 0.01$	Uninformative or upsampled crops
Longest phrases	Top 1% by length	Over-expansion, clause leakage
Shortest phrases	Single-word phrases	Generic phrases matching arbitrary regions
Most boxes per image	Top 1% by box count	Redundant boxes surviving NMS
Most elongated boxes	$a > 2.0$	Content lost by the center crop

Table 4.5. Strata of the manual review. Each stratum is tied to a distribution analyzed above and to the failure mode it probes.

cohort. Following the principle of inspecting edge cases that aggregate statistics obscure (Section 2.5), each stratum is explicitly tied to an empirical distribution feature analyzed above and a corresponding theoretical failure mode. Because these tail strata deliberately over-sample high-risk edge cases, aggregate metrics computed across all 180 items represent a conservative *stress-test lower bound* rather than an unweighted extrapolation to the whole dataset.

4.9.2 Annotation Scheme

For each sampled box, the full image, the box, and its phrase were shown together, and two kinds of judgments were recorded. The first is a single *correspondence label* for the region-phrase pair:

- **Correct**: the box contains the entity named by the phrase and is a reasonable localization of it.
- **Partially correct**: the box contains the entity, but is clearly too loose or too tight, or contains only a part of it. This category may contain plurality mismatch.
- **Incorrect**: the box localizes a different entity from the one described.
- **Ungroundable**: the phrase does not describe anything that can be localized, such as a remaining meta-descriptor or an abstract expression.

The second kind is a set of independent binary *diagnostic flags*, each of which is tied to a design decision of Chapter 3, so that an observed error can be traced back to the component responsible for it (Table 4.6).

Diagnostic flag	Design decision tested
Residual meta-descriptor	Syntactic-context filtering instead of the lexical blocklist (Section 3.2), including the acknowledged limitation for *frame*
Residual evaluative or commentary modifier	Stripping of deictic, evaluative, and commentary modifiers
Clause leakage or truncated phrase	Clause-boundary truncation and single-sentence isolation
Plurality / Granularity mismatch	Instance-level box resolution vs. multi-object group assignment and scale granularity
Compositional / Localization error	Choice of OWLv2 over deep cross-attention grounding (Section 3.3) for complex attribute/relation binding

Table 4.6. Diagnostic flags recorded during the manual review and the corresponding design decisions tested.

4.9.3 Reliability of the Labels

A manual judgment is only meaningful if it is reproducible. Inter-rater agreement in nominal annotation tasks is typically quantified using Cohen's κ [36]. The review in this thesis was carried out by a single reviewer (the author); potential single-annotator bias is noted as a limitation in Section 4.10.

4.9.4 Results

Overall precision and stress-test performance: On the unbiased random baseline sample ($n_s = 20$), 60.0% of the region-phrase pairs were classified as strictly correct (95% Wilson confidence interval [38.7%, 78.5%]), 35.0% exhibited minor label noise, 0.0% were incorrect, and 5.0% were ungroundable. Under a lenient evaluation criterion that admits minor label noise (such as single-object plurality mismatches or sub-optimal bounding box cropping), baseline random precision reaches 95.0% ([76.4%, 99.1%]). Across the entire audited pool ($N = 180$), strict accuracy is 57.8% ([50.4%, 64.9%]) and lenient accuracy is 87.8% ([82.2%, 91.8%]). Crucially, as established in the sampling design, because 88.9% of the audited cohort ($n = 160$) was deliberately drawn from failure-prone tail strata, these aggregate metrics across $N = 180$ represent a conservative *stress-test lower bound* under severe geometric and structural edge cases, rather than an unweighted estimate of global dataset precision. Table 4.7 details the complete breakdown of region-phrase correspondence per stratum.

Embedded text and sign localization errors: A distinct, recurring failure mode observed during the manual audit involved candidate phrases referencing embedded or quoted text within the visual scene (e.g., street signs, store names, textual logos, or printed background labels). The open-vocabulary detector (OWLv2) [28] consistently struggled with precise spatial localization for these samples. These failures manifested in two primary ways: *partial localization*, where the bounding box captured only a fraction of the sign or word, and *mislocalization*, where the detector assigned the text phrase to an adjacent non-text region or object frame. This highlights a fundamental limitation of general-purpose vision-language detection backbones when performing fine-grained text-anchored grounding without dedicated optical character recognition (OCR) or text-spotting modules [37].

Stratum	n	Correct (%)	Part. correct (%)	Incorrect (%)	Ungroundable (%)
Random	20	60.0%	35.0%	0.0%	5.0%
Lowest confidence	20	65.0%	35.0%	0.0%	0.0%
Highest confidence	20	70.0%	20.0%	10.0%	0.0%
Largest boxes	20	70.0%	15.0%	10.0%	5.0%
Smallest boxes	20	85.0%	0.0%	15.0%	0.0%
Longest phrases	20	50.0%	15.0%	35.0%	0.0%
Shortest phrases	20	40.0%	45.0%	15.0%	0.0%
Most boxes per image	20	50.0%	50.0%	0.0%	0.0%
Most elongated boxes	20	80.0%	10.0%	10.0%	0.0%

Table 4.7. Correspondence labels per stratum of the manual review ($N = 180$). Correct represents strict alignment; partially correct includes minor noise such as plurality/granularity discrepancies.

Diagnostic flag	Random sample (%)	Probing stratum (%)
Residual meta-descriptor	0.0%	5.0%
Residual evaluative or commentary modifier	0.0%	5.0%
Clause leakage or truncated phrase	0.0%	10.0%
Plurality / Granularity mismatch	35.0%	50.0%
Compositional / Localization error	0.0%	35.0%

Table 4.8. Frequency of diagnostic flags. The probing column reports frequencies in the corresponding risk-targeted strata (e.g., longest phrases, most boxes per image).

Information gained from confidence scores: Comparing the lowest-confidence stratum (detector score 0.30–0.40) against the highest-confidence stratum (detector score > 0.70), the share of strictly correct pairs increases from 65.0% to 70.0%, whereas hard failures (incorrect pairs) increase from 0.0% to 10.0%. While detector confidence provides a moderate ordinal signal for strict spatial fit, it fails to reliably eliminate minor text-side noise. Consequently, raw confidence scores should be interpreted as an ordinal heuristic rather than an absolute proxy for visual-textual quality.

Table 4.8 reports the frequency of diagnostic flags across the random baseline and targeted probing strata designed to expose specific edge cases. Plurality mismatches (e.g., bounding a single isolated object under a plural noun phrase) constituted the single most frequent minor failure mode, occurring in 35.0% of random baseline samples and rising up to 50.0% within dense object regions. Conversely, head-noun ambiguities (such as *frame*, *border*, or ungroundable generic descriptors) were effectively suppressed by Stage 1 filtering, appearing in only 5.0% of the largest bounding box crops.

Figure 4.7 presents representative samples from the audit: strictly correct pairs, cases with minor label noise (such as single-item plurality assignment), and severe failure modes (such as truncated clause grounding or hallucinated entity bounds).

4.9.5 Effect of the Text-Side Filters: Comparison with a Naive Baseline

To evaluate the impact of single-sentence isolation, multi-layer filtering, and clause-boundary truncation on text-side quality, a point of comparison was estab-



Figure 4.7. Examples of region-phrase pairs from the dataset, illustrating correct grounding alongside primary minor noise and failure categories.

Measure (phrase level)	Naive baseline	Final pipeline
Phrases per caption	46.02	2.1
Ungroundable phrases (%)	1.52	2.2
Residual meta-descriptor flag (%)	4.44	0.6
Evaluative or commentary modifier flag (%)	0.8	1.1
Clause leakage flag (%)	7.8	1.7
Median phrase length (words)	2.0	2.3

Table 4.9. Text-side quality metrics comparing the naive extraction baseline against the final Stage 1 pipeline across $n_b = 180$ captions.

lished using a naive baseline. This baseline extracts standard spaCy noun chunks directly from full captions without applying sentence isolation, noise filtering, or syntactic boundary constraints. For $n_b = 180$ captions, candidate phrases from both approaches were evaluated prior to detector grounding. As shown in Table 4.9, the Stage 1 pipeline drastically reduces candidate density from 46.02 down to 2.1 phrases per caption, effectively filtering out over 95% of redundant, over-segmented text candidates. This structural pruning notably suppresses text-side noise, cutting clause leakage from 7.8% down to 1.7% and reducing residual meta-descriptors from 4.44% to 0.6%. The small increases in ungroundable phrases (1.52% to 2.2%) and evaluative modifiers (0.8% to 1.1%) are probably a consequence of the much smaller number of phrases per caption: once most noisy phrases are removed, a similar number of residual errors makes up a larger percentage. This was not verified on absolute counts. By eliminating over 43 noisy or trivial phrases per caption, the total candidate pool shrinks significantly, naturally concentrating the relative percentage of any surviving complex descriptors. Crucially, the median phrase length increases from 2.0 to 2.3 words, confirming that the remaining candidates retain crucial descriptive specificity while shedding non-entity clause clutter.

4.10 Limitations of the Analysis

The conclusions of this chapter are subject to several limitations that have to be kept in mind when interpreting them.

First, the analysis starts from the finished dataset. The number of candidates and detections after each filtering step was not recorded during the run, so this chapter cannot say which stage is responsible for most of the loss, nor whether the records dropped in Stage 3 differ systematically from the retained ones. In the same way, the manual review measures the *precision* of the retained pairs but not the *recall* of the grounding, i.e. how many groundable entities in the captions received no box. Both aspects would require instrumenting the pipeline, and are left as future work.

Second, the manual review is of limited size and relies on a single reviewer, making interpretations along the boundary between *correct* and *partially correct* inherently subjective. While Cohen's κ [36] is standard for assessing inter-rater agreement, single-annotator constraints precluded multi-rater reliability estimation; subjectivity was mitigated solely through strict annotation guidelines. Confidence intervals [35] quantify the sampling uncertainty.

Third, the area ratio and the confidence are, by themselves, no measure of correctness as a box can have a plausible size and a high score and still be wrong. They serve to locate suspicious regions of the data, which the manual review then examines.

4.11 Summary

Returning to the six questions of Section 4.1, the analyses lead to the following picture. **(Q1)** The final dataset contains 608,290 samples with 4,397,937 boxes. **(Q2)** A sample from the final dataset carries on average 7.23 boxes per image. **(Q3)** The text boxes have a median length of 3.00 words, which is a small fraction of the full caption length and fits comfortably within standard text encoder context windows [1]. **(Q4)** Across the analyzed dataset, the boxes cover a median of 0.0380 (3.8%) of the image area, with 1.33% near-full-image regions and 50.15% very small boxes. **(Q5)** The retained confidence scores are heavily concentrated near the 0.30 detection threshold, and raw detector scores alone do not reliably predict visual grounding precision. **(Q6)** In the manual review ($N = 180$), 87.8% of the grounded text-box pairs are visually correct under lenient criteria (57.8% under strict criteria), and the text-side filters reduce structural clause leakage from 7.8% down to 1.7% (and residual meta-descriptors from 4.44% down to 0.6%) compared with a naive extraction. The primary limitation observed is the heavy prevalence of tiny background box clutter (50.15%), pointing directly to detector over-detection in Stage 2 and highlighting the need for stricter spatial scale filtering in future iterations.

Chapter 5

Conclusion and Future Work

5.1 Summary of Main Findings

This thesis presented an automated four-stage pipeline for constructing a densely grounded vision-language dataset from PixMo-Cap's detailed visual narrations [22]. Through dataset validation combining corpus-wide statistical analysis and stratified manual reviews, the evaluation confirmed the visual and textual quality of the resulting corpus:

- **Text-Side Filtering Efficiency:** Stage 1 effectively mitigated text-side noise inherent in raw dense captions. Compared to a naive spaCy baseline [24], candidate phrase density was streamlined from 46.02 to 2.1 phrases per caption, reducing structural clause leakage from 7.8% down to 1.7% and residual meta-descriptors from 4.44% down to 0.6%.
- **Dense Spatial Grounding:** The final dataset yields 608,290 samples containing 4,397,937 grounded bounding boxes (averaging 7.23 boxes per sample), providing a significantly richer multi-region hierarchy than reference datasets such as GRIT (1.5 boxes/image) [11].
- **Text-Visual Alignment Quality:** In the manual review of 180 samples, 87.8% of the region–phrase pairs were judged correct under the lenient criterion and 57.8% under the strict one. Because 160 of these samples were drawn from failure-prone strata, these figures are a lower bound and not an estimate for the whole dataset (Section 4.9).

5.2 Discussion, Lessons Learned, and Limitations

5.2.1 What Worked and What Did Not

The combination of single-sentence isolation, dependency-tree expansion with a preposition whitelist, and syntactic modifier stripping succeeded in generating informative, non-clause-leaked text spans. Leveraging OWLv2 [28] enabled fast, open-vocabulary spatial grounding across diverse visual concepts. However, raw detector confidence scores served primarily as an ordinal heuristic ($c \geq 0.30$) rather than a linear predictor of visual precision.

Furthermore, the pipeline exhibited two prominent grounding failures:

1. **Micro-Background Clutter:** An overabundance of tiny background boxes occurred, with 50.15% of all retained boxes covering less than 1.0% of the total image area, revealing detector over-sensitivity in Stage 2.

2. **Embedded Text and Sign Mislocalization:** Candidate phrases describing quoted text, signage, store names, or written labels constituted the majority of hard localization errors. OWLv2 [28] frequently produced partial bounding boxes (truncating written words) or completely mislocated text-based visual entities, underscoring the gap between general open-vocabulary object detection and fine-grained scene text spotting [37].

5.2.2 Dataset Differences and Preprocessing Challenges

Unlike GRIT's brief web-scraped image-text pairs [11], PixMo-Cap consists of long, narration-style captions (averaging 196 words) [22]. Without Stage 1's clause-boundary truncation, raw extraction collapsed into long, clause-heavy sentences. On the spatial side, frequent plurality mismatches (35% in random samples) occurred when single object instances were bounded under plural phrase queries, highlighting the difficulty of handling group-level versus instance-level spatial grounding automatically.

5.3 Future Work

Several avenues remain open for improving dataset construction pipelines and extending downstream applications:

- **Pipeline Instrumentation and Recall Tracking:** Future pipeline iterations should log candidate counts at each processing stage to evaluate exact stage-by-stage drop rates and quantify text grounding recall.

- **Dedicated OCR and Sign Grounding:** To resolve the high rate of partial bounding and mislocalization on quoted texts and signs, future pipelines could integrate a dedicated OCR/text-spotting module to extract and ground image-embedded text prior to standard region candidate matching [37].

- **Scale and Plurality Filtering:** Introducing an explicit minimum bounding box area threshold during Stage 3 will reduce micro-background clutter. Additionally, integrating grammatical number resolution will eliminate single-versus-plural instance assignment mismatches.

- **HyCoCLIP Model Pretraining:** Future research can explore training HyCoCLIP [10] directly on this new dataset, investigating how hyperbolic geometric embeddings leverage high-density region-phrase hierarchies to organize complex visual concepts and fine-grained spatial relationships.

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